

Supplemental Material for “Basis-Dependent Sparsity and Energy Recovery in a Strongly Correlated [2Fe–2S] Active Space”

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This Supplemental Material defines the active-space Hamiltonian, gives the complete staged Davidson simulation, documents direct and density-matrix replay, derives and validates the natural-orbital transformation, specifies the top- K and Hamiltonian-decomposition experiments, and maps every result to an executable script and frozen artifact. “Full configuration interaction” below always means the complete determinant space of the specified CAS(30e, 20o) Hamiltonian in the fixed $M_S = 0$ sector; it does not mean full-molecule or complete-basis FCI.

HAMILTONIAN, DETERMINANT CONVENTION, AND PUBLIC INPUTS

The calculation uses a real, spin-independent FCIDUMP with 20 orthonormal spatial orbitals, 30 electrons, $M_S = 0$, and $E_{\text{core}} = 0$. In the chemist’s integral convention implemented by PySCF,

$$\hat{H} = \sum_{pq\sigma} h_{pq} a_{p\sigma}^\dagger a_{q\sigma} + \frac{1}{2} \sum_{pqrs\sigma\tau} (pq|rs) a_{p\sigma}^\dagger a_{r\tau}^\dagger a_{s\tau} a_{q\sigma}. \quad (1)$$

There are 15 alpha and 15 beta electrons. Each spin sector contains $\binom{20}{15} = 15,504$ occupation strings, so the CI tensor has shape $15,504 \times 15,504$, dimension 240,374,016, and occupies 1,922,992,256 bytes in little-endian float64 NPY format.

The primary FCIDUMP is retrieved from the immutable content associated with Ref. [1]; its SHA-256 is

bd6ef31773f7b082171f73faa6c98924948e2c250b6fbec74504d64361c8c6c7.

The second, independently distributed QMB FCIDUMP is retrieved at repository revision ab1a7002ccbaca0ac43acafaf0f40400d14c3940 from Ref. [2]; the uncompressed file has SHA-256

95d8786af06eeea2107e19ffd98c66a6ca97fc8c9864175a4f6d64512b6f2df9.

Orbital equivalence of the two FCIDUMPs

The two files are not byte-identical because their active orbitals differ. The script `code/compare_hamiltonian_orbital_equivalence.py` diagonalizes each real symmetric one-electron matrix. Its spectrum is nondegenerate, with a minimum adjacent gap of $1.6042 \times 10^{-2} E_h$, so the ordered eigenvectors are unique up to column signs. After fixing the first sign and determining the remaining signs from nonzero two-electron witnesses, the orthogonal matrix U is tested against

$$h_{pq}^{(B)} = \sum_{ij} U_{ip} h_{ij}^{(A)} U_{jq}, \quad (2)$$

$$g_{pqrs}^{(B)} = \sum_{ijkl} U_{ip} U_{jq} U_{kr} U_{ls} g_{ijkl}^{(A)}. \quad (3)$$

The final comparison contracts the complete two-electron tensor without screening. Table I establishes orbital-rotation equivalence to the numerical precision of the inputs; it does not attach an uncertainty to a rounded literature energy.

CHECKPOINTED DAVIDSON SIMULATION

Matrix-free Ritz iteration

The 240,374,016-dimensional Hamiltonian matrix is never formed. PySCF’s `fci.direct_spin1` implementation applies the Slater–Condon Hamiltonian to a CI tensor. At update m , an orthonormal trial basis V_m defines

$$T_m = V_m^\text{T} H V_m. \quad (4)$$

TABLE I. Orbital-equivalence checks. “Maximum” denotes the largest elementwise absolute difference.

Quantity	Value
Maximum ordered h -eigenvalue difference	$2.1316 \times 10^{-14} E_h$
Maximum defect of $U^T U = I$	3.1086×10^{-15}
Maximum rotated one-electron residual	$3.5527 \times 10^{-14} E_h$
RMS complete two-electron-tensor residual	$3.0805 \times 10^{-15} E_h$
Maximum complete two-electron-tensor residual	$1.2055 \times 10^{-13} E_h$

TABLE II. Davidson stages. Energies and residuals are direct contractions of the saved endpoint. The update limit is a numerical stop, not a convergence claim.

Stage	Updates	Space	Energy (E_h)	Residual (E_h)	E tol.	Vector SHA-256 prefix
A: initialization	110	14	-116.6055095343891	2.8201×10^{-3}	10^{-10}	3b20ec2e1aa3
B1: wide restart	2	64	-116.6055107620259	2.0814×10^{-3}	10^{-13}	cc9350d4bdad
B2: checkpoint	16	64	-116.6055790186671	4.1575×10^{-3}	10^{-13}	1664c06a55c7
C: production	160	64	-116.60560912042631	6.9016×10^{-7}	10^{-13}	45e63bb63cd1

The lowest eigenpair (θ_m, y_m) of T_m gives the Ritz vector $c_m = V_m y_m$ and residual $r_m = H c_m - \theta_m c_m$. A diagonally preconditioned correction expands the basis; the iteration restarts when the configured subspace capacity is reached. We set `davidson_only=True` and `pspace_size=800` so the reported path is iterative throughout.

The implementation in `code/continue_fci_checkpointed.py` adds three reproducibility safeguards. First, it hashes the FCIDUMP and input vector before numerical work. Second, every eighth callback writes the current Ritz vector to a temporary NPY, flushes and synchronizes it, and atomically renames it. Third, after each stage it reloads the saved endpoint, constructs Hc , and recomputes the norm, Rayleigh quotient, and residual. Consequently, every reported endpoint is an observable of a concrete coefficient tensor rather than an extrapolation or a printed scalar.

Complete staged protocol

Table II gives every stage used to obtain the reported state. Stage A starts from PySCF’s default determinant-space initial guess. B1 tests and establishes a wider 64-vector restart. B2 creates the synchronized input for the 160-update production stage C. Only the explicit vector listed for a stage is passed to the next one.

Stages B and C used the residual target $10^{-8} E_h$, 64 software threads, a 512-GiB node allocation, and reported maximum memory 450,000 MB. The production eigensolver wall time was 23,387.508 s. The complete driver, including initial and final contractions, checkpoint input/output, and hashing, took 23,494.156 s. The final solver flag is `converged=false` : the 160-update limit was exhausted while the residual remained above its target.

The final coefficient tensor is `fci_ritz_vector.npy` , with SHA-256

45e63bb63cd10e953260b6f5dabb8a1dda5518fbf5959002c23888dacb599945.

Every downstream script asserts this hash, the FCIDUMP hash, the tensor shape, and float64 dtype before evaluating a reported quantity.

FRESH-PROCESS REPLAY AND NUMERICAL STABILITY

The script `code/verify_publication_state.py` starts in a fresh Python process and does not import the optimization driver. It evaluates the energy by two paths. The first absorbs the one-electron integrals into the two-electron contraction and explicitly constructs $H_{el}c$:

$$N = \langle c | c \rangle, \quad (5)$$

$$E_{Hc} = E_{\text{core}} + \frac{\langle c | H_{el} | c \rangle}{N}, \quad (6)$$

$$\rho = \frac{\|H_{el}c - (E_{Hc} - E_{\text{core}})c\|_2}{\sqrt{N}}. \quad (7)$$

The second constructs separate one- and two-particle reduced density matrices and evaluates

$$E_{\text{RDM}} = E_{\text{core}} + \sum_{pq} h_{pq} \gamma_{qp} + \frac{1}{2} \sum_{pqrs} g_{pqrs} \Gamma_{pqrs}. \quad (8)$$

Both paths use PySCF, but they have distinct contraction routes; “replay” does not mean a second electronic-structure library. Results appear in Table III.

TABLE III. Fresh-process observables of the saved state.

Quantity	Value
Squared norm	0.9999999999999999
Direct Rayleigh energy	$-116.60560912042631 E_h$
RDM-reconstructed energy	$-116.60560912042571 E_h$
Absolute energy-path difference	$5.9686 \times 10^{-13} E_h$
Residual norm	$6.9016 \times 10^{-7} E_h$
Energy variance ρ^2	$4.7631 \times 10^{-13} E_h^2$
$\langle \hat{S}^2 \rangle$	2.8570×10^{-9}
Spin multiplicity	1.0000000057

Two PySCF kernels also accumulated the spin-summed one-particle RDM through different paths. On the 240-million-component tensor their maximum elementwise difference was 1.7360×10^{-10} , at diagonal element (16, 16), and therefore exceeded an initially chosen 10^{-10} absolute comparison. The subsequent stability calculation found a relative Frobenius difference 8.4093×10^{-11} , zero reported symmetry defect in either RDM, and traces 29.999999997413795 and 29.99999999999968. Equation (8) retained sub-pico-Hartree agreement with the direct energy. We report this as an accumulation-order stability study rather than replacing the initial comparison by a looser undeclared threshold.

EXACT NATURAL-ORBITAL REPRESENTATION OF THE SAME STATE

The real symmetric, spin-summed one-particle RDM is symmetrized and diagonalized,

$$\frac{\gamma + \gamma^T}{2} U = U \text{diag}(n_1, \dots, n_{20}), \quad U^T U = I. \quad (9)$$

The natural occupations sum to 29.9999999974. Ten values are strongly fractional, ranging from 0.604296 to 1.468175; the complete ordered list is stored in `data/natural_occupations.csv`. The maximum off-diagonal RDM element after rotation is 1.07×10^{-15} , and the maximum orthogonality defect of U is 2.44×10^{-15} .

The integral tensors are transformed with the same U according to Eqs. (2) and (3). The determinant transformation for one spin sector is the exterior-power representation of U : a matrix element between two occupation strings is the corresponding 15-by-15 minor. Direct evaluation of all such minors is unnecessarily expensive because the sector contains only five holes. Jacobi’s complementary-minor identity,

$$\det U_{I,J} = (-1)^{\Sigma I + \Sigma J} \det(U) \det[(U^{-1})_{J^c, I^c}], \quad (10)$$

maps every 15-electron minor to a 5-hole minor. The script `analysis/transform_ci_natural_basis.py` maps the electron tensor to the hole ordering, calls PySCF’s determinant transformation in the five-hole sector, restores electron ordering and parity, and writes the transformed float64 tensor. A seeded six-orbital, four-electron test compares the direct electron and complementary-minor paths and gives maximum difference 2.50×10^{-16} .

The source and transformed tensors are contracted in the same hash-locked process before any sparsity analysis. Table IV shows that the transformation changes representation but not the physical state. The natural-basis FCIDUMP, rotation, and transformed vector have SHA-256 values `adb401872023...`, `3a3dcdbdfac4c...`, and `3855dc7c7ce8c...`, respectively; their full hashes are recorded in the manifest.

An independent frozen-vector calculation further tests the direction of the orbital transformation, rather than only its spectrum. Before reading either FCIDUMP or CI tensor, it verifies the SHA-256 of the benchmark-basis occupation list, `2e9632ff525337eae6329eac5cd5e39b71bd08af85291a335033296b5a732b7d`. It then constructs the one-particle RDM directly in the natural representation and compares the complete symmetrized matrix elementwise with the diagonal matrix made from that frozen list. The final trace is 29.999999987374895, compared with $N\langle c|c \rangle =$

TABLE IV. Same-process orbital-transformation checks.

Quantity	Benchmark basis	Natural basis	Difference
Squared norm	0.9999999999999999	0.9999999999999985	1.39×10^{-14}
Energy (E_h)	-116.60560912042631	-116.60560912042600	3.13×10^{-13}
Residual (E_h)	$6.9015577054 \times 10^{-7}$	$6.9015577065 \times 10^{-7}$	1.12×10^{-16}

30.000000000000426; the absolute difference is 1.26×10^{-8} . The maximum raw symmetry defect, off-diagonal element, complete-matrix difference, and occupation-eigenvalue difference are 0, 2.27×10^{-11} , 6.27×10^{-10} , and 6.27×10^{-10} , respectively.

A preliminary replay used an absolute trace threshold of 10^{-8} and returned a trace of 29.999999987374895, approximately 1.26×10^{-8} from $N\langle c|c\rangle$. Before the final replay, we fixed a 2×10^{-8} absolute tolerance and added the stronger direction-sensitive complete-matrix comparison above; neither threshold nor test was altered after observing the final output. This records the accumulation-level numerical tolerance without converting it into a physical uncertainty on the energy.

TOP-K PROJECTIONS AND EXACT DECOMPOSITION

For each orbital representation, the streaming ranking in `analysis/analyze_state_compressibility.py` retains the exact global largest 4,194,304 values of $|c_I|^2$. At each input block it keeps the block's local candidates, merges them with the current global set, and applies an exact partition; a final stable sort fixes descending order. The 17 support sizes are

$$K = 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, \\ 65536, 131072, 262144, 524288, 1048576, 2097152, 4194304. \quad (11)$$

For every K , a zeroed full-shaped tensor receives the selected coefficients without reoptimization and is normalized. A fresh $H\phi_K$ contraction gives its Rayleigh energy and full-space residual. The program requires nested cumulative norm and $E_K \geq E_\star - 10^{-10} E_h$, and exports all points as CSV. This protocol never converts norm loss into an assumed energy loss.

For the benchmark basis, the coefficient probabilities $p_I = |c_I|^2$ give

$$N_{\text{IPR}} = \left(\sum_I p_I^2\right)^{-1} = 5525.89, \quad N_{\text{Shannon}} = \exp\left[-\sum_I p_I \ln p_I\right] = 109228.69. \quad (12)$$

These scalar counts are representation dependent. The direct two-basis comparison is summarized in Fig. 2 and the complete files `data/topk_tracker_basis.csv` and `data/topk_natural_basis.csv`. At $K = 4,194,304$, the natural-orbital projection increases W_K from 0.985678 to 0.999776 and reduces ΔE_K from 17.705 to 0.3488 m E_h ; its residual nevertheless remains $2.72 \times 10^{-2} E_h$.

TABLE V. Representative amplitude-ranked projections. Energy errors are relative to the full Ritz state and every residual is evaluated in the complete active-space determinant space.

K	Benchmark orbitals			Natural orbitals		
	W_K	ΔE_K (m E_h)	residual (E_h)	W_K	ΔE_K (m E_h)	residual (E_h)
64	0.075498	784.161	0.840786	0.418118	406.615	0.688659
512	0.191346	605.266	0.785570	0.768507	168.983	0.472888
65,536	0.726934	212.378	0.546320	0.974209	25.545	0.207036
4,194,304	0.985678	17.705	0.186547	0.999776	0.3488	0.027168

To decompose the error invariantly under an additive Hamiltonian constant, set $A = H - E_\star I$, $p = P_K c$, $q = (I - P_K)c$, and $W = \langle p|p\rangle$. Because $c = p + q$ and $\langle c|A|c\rangle = 0$,

$$W(E_K - E_\star) = \langle p|A|p\rangle \quad (13)$$

$$= -\langle q|A|q\rangle - 2\text{Re}\langle p|A|q\rangle. \quad (14)$$

The analysis contracts $H\phi_K$. With $p = \sqrt{W}\phi_K$, it obtains $\langle p|H|p \rangle$ and $\langle p|H|c \rangle$, then reconstructs $2\text{Re}\langle p|H|q \rangle$ by subtraction and $\langle q|H|q \rangle$ from the complete-state energy identity. Thus the tail term is an algebraic reconstruction, not a separate Hq contraction. Across both orbital bases and all 17 support sizes, the largest floating-point discrepancy on recombining Eq. (14) is $6.1 \times 10^{-14} E_h$. The positive removed-coupling and negative tail-quadratic contributions in Fig. 2 are $-2\text{Re}\langle p|A|q \rangle/W$ and $-\langle q|A|q \rangle/W$, respectively.

HAMILTONIAN-AWARE 512-DETERMINANT CONTROLS

Five fixed-budget states separate inherited coefficients, coefficient relaxation, orbital representation, and support choice. In each orbital representation, the top-512 amplitude support and inherited coefficients are taken directly from the full Ritz tensor. The script `analysis/rediagonalize_topk_support.py` then builds the complete 512-by-512 Hamiltonian on exactly that support and diagonalizes it. In the benchmark representation, a separate selected-CI calculation constructs a different support: it grows an approximately 80,000-determinant candidate space by Hamiltonian connections and Epstein–Nesbet estimates, ranks additions by secular energy lowering, evaluates removals by the exact Cauchy secular equation, accepts only energy-lowering swaps, and rediagonalizes every accepted 512-dimensional support. Because the search uses seeded, finite-time large-neighborhood restarts, this state is a reproducible candidate, not a claim of a unique or globally optimal support.

For all three rediagonalized states, the dense subspace Hamiltonian is built by `analysis/reference_slater_condo_n.py`, a compact implementation that parses the FCIDUMP and applies explicit fermionic creation and annihilation operators. A separate PySCF embedding places the 512 coefficients into the full $15,504^2$ tensor and contracts Hc . Table VI reports the results. The projected residual tests stationarity only within the listed support; the full-space residual tests coupling to all excluded determinants.

TABLE VI. Fixed-budget 512-determinant controls. Energy error is relative to E_* . “Projected” and “full” denote residual norms in the selected subspace and complete active-space determinant space. B and N denote the benchmark and natural-orbital representations.

State	Energy (E_h)	Error (mE_h)	Projected residual (E_h)	Full residual (E_h)
B: amplitude support, inherited coefficients	-116.000343445108	605.266	—	0.786
B: same support, rediagonalized coefficients	-116.146308570453	459.301	2.42×10^{-13}	0.558
B: Hamiltonian-guided support, rediagonalized	-116.370462675845	235.146	8.68×10^{-14}	0.215
N: amplitude support, inherited coefficients	-116.436626144441	168.983	—	0.473
N: same support, rediagonalized coefficients	-116.486260278799	119.349	1.68×10^{-13}	0.311254

The same-support energy from the explicit fermionic implementation and PySCF differs by $3.13 \times 10^{-13} E_h$ in B and $5.68 \times 10^{-14} E_h$ in N; the corresponding difference for the Hamiltonian-guided state is $1.71 \times 10^{-13} E_h$. The frozen wave-function CSVs have 512 unique rows and squared norms equal to unity within 4×10^{-16} . The construction and validation codes are distinct, and the reported energy is always the independently contracted value.

SOFTWARE ENVIRONMENT AND REPRODUCTION MAP

The production state used Python 3.12.1, NumPy 2.5.1, SciPy 1.18.0, h5py 3.14.0, and PySCF 2.14.0. The PySCF and h5py wheel SHA-256 values are

```
37b0bccc55450311a55318cd643e851353331ddeab4fc0c0065e83c905e41502
0cbd41f4e3761f150aa5b662df991868ca533872c95467216f2bec5fcad84882.
```

Thread controls `OMP_NUM_THREADS`, `MKL_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, and `NUMEXPR_NUM_THREADS` were set to 64. The top-level `README.md` gives copy-paste commands for every stage and records expected hashes at each boundary.

The compact reproduction bundle includes source code, job manifests, environment locks, convergence history, verification JSON, the derived natural-orbital FCIDUMP and rotation, all figure data, and SHA-256 manifests; immutable locations and hashes for both public source FCIDUMPs are given above. For anonymous scientific review the 1,922,992,256-byte final vector is supplied as the separate artifact `fci_ritz_vector.npy` and identified by its full hash above. A persistent, unauthenticated public accession and final author metadata must replace the review-only transfer before external submission.

TABLE VII. Script-to-result map in the reproduction bundle.

Script	Principal inputs	Principal outputs
<code>code/initialize_fci_ritz.py</code>	Primary FCIDUMP	Stage-A vector and direct diagnostics
<code>code/continue_fci_checkpointed.py</code>	FCIDUMP, explicit NPY checkpoint	Atomic checkpoints, history JSONL, endpoint diagnostics
<code>code/verify_publication_state.py</code>	FCIDUMP, final vector	Direct/RDM energy, residual, norm, spin, occupations
<code>code/compare_hamiltonian_orbital_equality.py</code>	Two public FCIDUMPs	Orbital rotation and complete-tensor residuals
<code>analysis/analyze_state_compressibility.py</code>	FCIDUMP, representation-specific vector	Top- K CSV, decomposition CSV, occupations, summary JSON
<code>analysis/transform_ci_natural_basis.py</code>	Both FCIDUMPs, rotation, benchmark vector	Natural-basis vector and invariance JSON
<code>analysis/run_natural_basis_replay_v3.sh</code>	Natural FCIDUMP/vector, frozen occupations	Hash-locked two-basis replay and natural top-512 control
<code>analysis/rediagonalize_topk_support.py</code>	FCIDUMP, full vector, explicit reference code	Same-support ground state and two-path validation
<code>analysis/guided_512/exp_structure.py, opt3.py</code>	FCIDUMP and fixed search parameters	Hamiltonian-guided candidate pool and 512-state CSV
<code>analysis/validate_optimized_512.py</code>	FCIDUMP, guided 512 state, explicit reference code	Guided-support energy and residuals
<code>analysis/build_results_tex.py</code>	Gated compact JSON/CSV	Numerical macros used by both documents
<code>figures/build_figures.py</code>	Frozen compact CSV/JSON only	PDF, SVG, and 450-dpi PNG figures

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- [1] Quantum Advantage Tracker Contributors, [\[2Fe-2S\] \(30e,20o\) active-space FCIDUMP](#), Public benchmark data repository (2026), accessed August 9, 2026; the input hash is reported in the Supplemental Material.
- [2] USTC Knowledge Computing Laboratory, [QMB Models: \[2Fe-2S\] \(30e,20o\) FCIDUMP](#), Hugging Face dataset repository (2025), repository revision `ab1a7002ccbaca0ac43acafaf0f40400d14c3940`; uncompressed SHA-256 reported in the Supplemental Material.